# Hugging Face LLM Course, Chapters 1–7

## Completion Evidence — Jacob Lacy (Hugging Face Username: aavk)

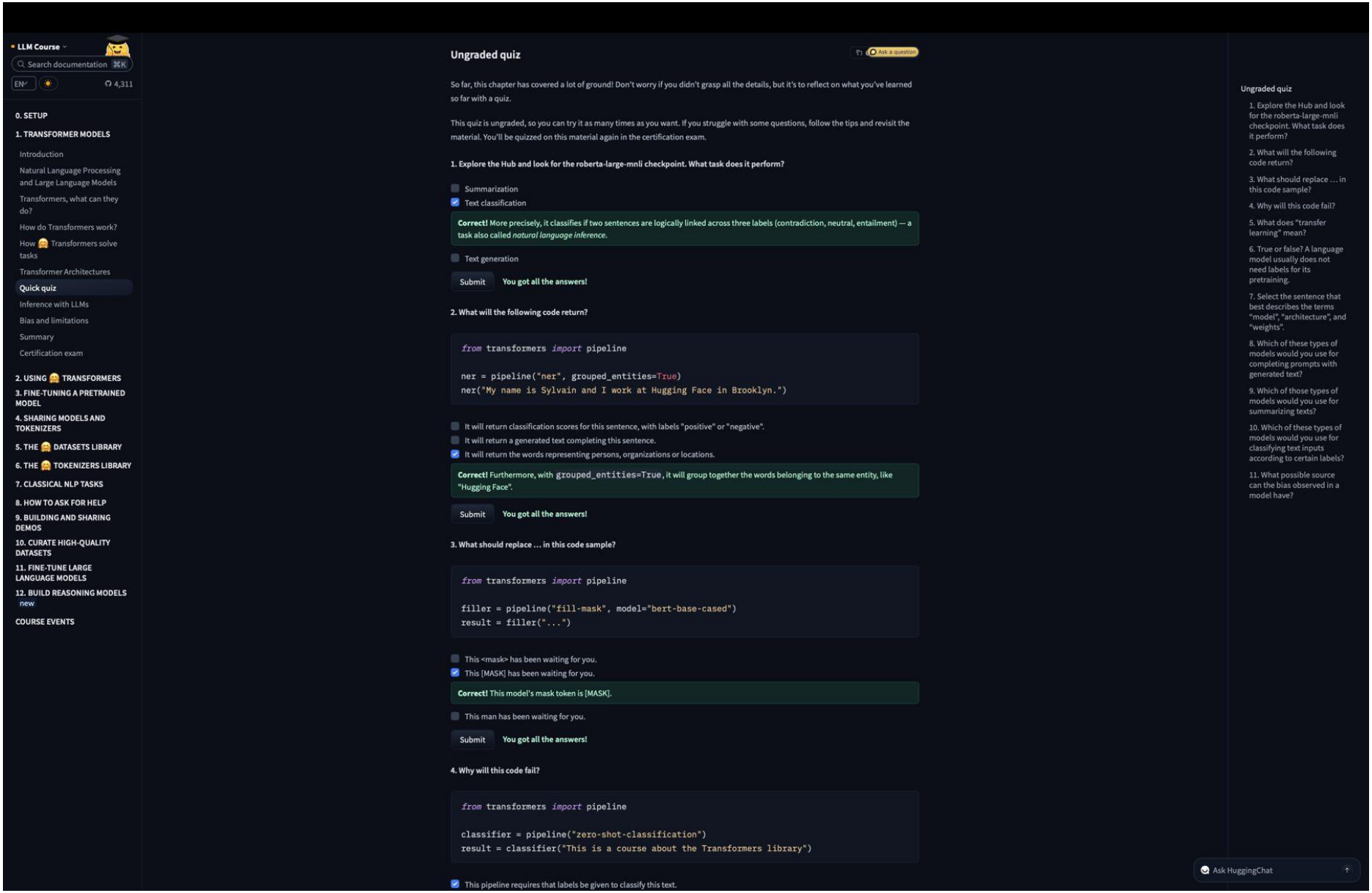
TRM 1B - Artefact A7

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The course has no overall certificate so completion is shown through the end-of-chapter quizzes for Chapters 1–7 and the Chapter 1 and Chapter 3 certification exams.

# Chapter 1 – Transformer Models: Quiz



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**5. What does "transfer learning" mean?**

- [ ] Transferring the knowledge of a pretrained model to a new model by training it on the same dataset.
- [x] Transferring the knowledge of a pretrained model to a new model by initializing the second model with the first model's weights.

**Correct!** When the second model is trained on a new task, it *transfers* the knowledge of the first model.

- [ ] Transferring the knowledge of a pretrained model to a new model by building the second model with the same architecture as the first model.

Submit **You got all the answers!**

**6. True or false? A language model usually does not need labels for its pretraining.**

- [x] True

**Correct!** The pretraining is usually *self-supervised*, which means the labels are created automatically from the inputs (like predicting the next word or filling in some masked words).

- [ ] False

Submit **You got all the answers!**

**7. Select the sentence that best describes the terms "model", "architecture", and "weights".**

- [ ] If a model is a building, its architecture is the blueprint and the weights are the people living inside.
- [ ] An architecture is a map to build a model and its weights are the cities represented on the map.
- [x] An architecture is a succession of mathematical functions to build a model and its weights are those functions parameters.

**Correct!** The same set of mathematical functions (architecture) can be used to build different models by using different parameters (weights).

Submit **You got all the answers!**

**8. Which of these types of models would you use for completing prompts with generated text?**

- [ ] An encoder model
- [x] A decoder model

**Correct!** Decoder models are perfectly suited for text generation from a prompt.

- [ ] A sequence-to-sequence model

Submit **You got all the answers!**

**9. Which of those types of models would you use for summarizing texts?**

- [ ] An encoder model
- [ ] A decoder model
- [x] A sequence-to-sequence model

**Correct!** Sequence-to-sequence models are perfectly suited for a summarization task.

Submit **You got all the answers!**

**10. Which of these types of models would you use for classifying text inputs according to certain labels?**

- [x] An encoder model

**Correct!** An encoder model generates a representation of the whole sentence which is perfectly suited for a task like classification.

- [ ] A decoder model
- [ ] A sequence-to-sequence model

Submit **You got all the answers!**

**11. What possible source can the bias observed in a model have?**

Ungraded quiz
1. Explore the Hub and look for the roberta-large-mnli checkpoint. What task does it perform?
2. What will the following code return?
3. What should replace ... in this code sample?
4. Why will this code fail?
5. What does "transfer learning" mean?
6. True or false? A language model usually does not need labels for its pretraining.
7. Select the sentence that best describes the terms "model", "architecture", and "weights".
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9. Which of those types of models would you use for summarizing texts?
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**11. What possible source can the bias observed in a model have?**

- [x] The model is a fine-tuned version of a pretrained model and it picked up its bias from it.

**Correct!** When applying Transfer Learning, the bias in the pretrained model used persists in the fine-tuned model.

- [x] The data the model was trained on is biased.

**Correct!** This is the most obvious source of bias, but not the only one.

- [x] The metric the model was optimizing for is biased.

**Correct!** A less obvious source of bias is the way the model is trained. Your model will blindly optimize for whatever metric you chose, without any second thoughts.

Submit **You got all the answers!**

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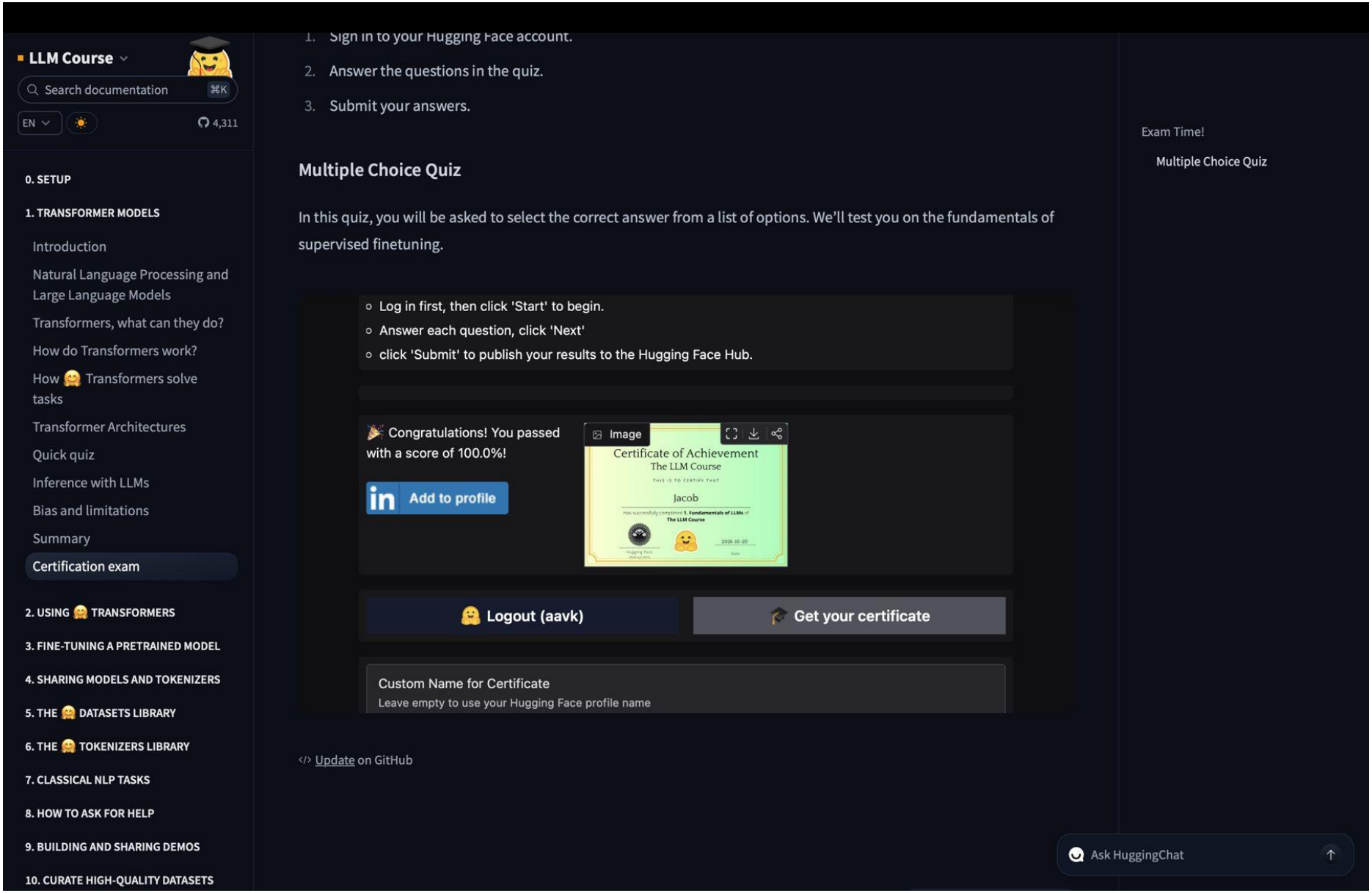
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## Chapter 1 – Certification Exam (100%) and Certificate



# Certificate of Achievement

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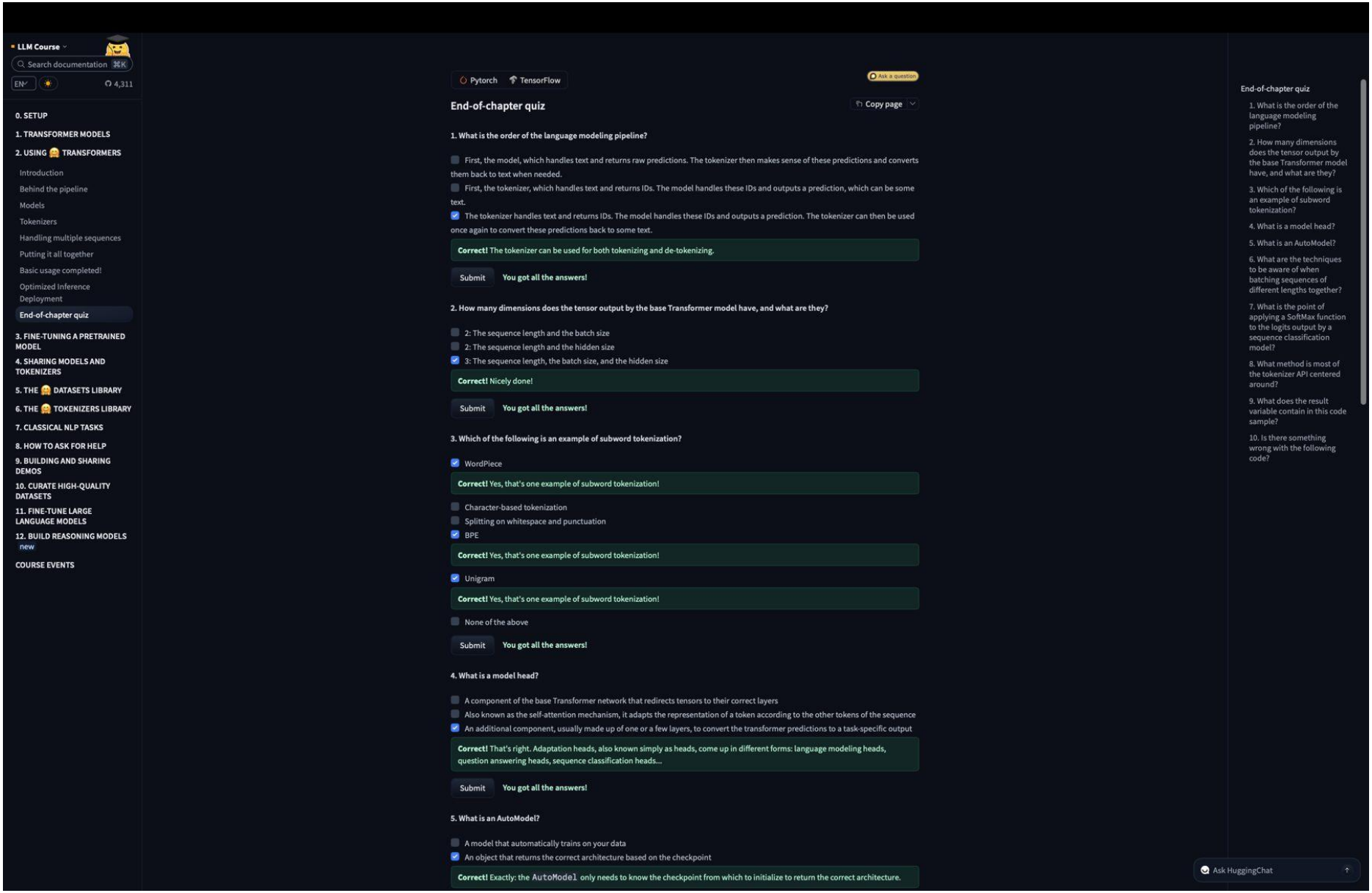
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# Chapter 2 – Using Transformers: Quiz



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**6. What are the techniques to be aware of when batching sequences of different lengths together?**

- [x] Truncating

**Correct!** Yes, truncation is a correct way of evening out sequences so that they fit in a rectangular shape. Is it the only one, though?

- [ ] Returning tensors
- [x] Padding

**Correct!** Yes, padding is a correct way of evening out sequences so that they fit in a rectangular shape. Is it the only one, though?

- [x] Attention masking

**Correct!** Absolutely! Attention masks are of prime importance when handling sequences of different lengths. That's not the only technique to be aware of, however.

Submit **You got all the answers!**

**7. What is the point of applying a SoftMax function to the logits output by a sequence classification model?**

- [ ] It softens the logits so that they're more reliable.
- [x] It applies a lower and upper bound so that they're understandable.

**Correct!** The resulting values are bound between 0 and 1. That's not the only reason we use a SoftMax function, though.

- [x] The total sum of the output is then 1, resulting in a possible probabilistic interpretation.

**Correct!** Correct! That's not the only reason we use a SoftMax function, though.

Submit **You got all the answers!**

**8. What method is most of the tokenizer API centered around?**

- [ ] `encode`, as it can encode text into IDs and IDs into predictions
- [x] Calling the tokenizer object directly.

**Correct!** Exactly! The `__call__` method of the tokenizer is a very powerful method which can handle pretty much anything. It is also the method used to retrieve predictions from a model.

- [ ] `pad`
- [ ] `tokenize`

Submit **You got all the answers!**

**9. What does the result variable contain in this code sample?**

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
result = tokenizer.tokenize("Hello!")
```

- [x] A list of strings, each string being a token

**Correct!** Absolutely! Convert this to IDs, and send them to a model!

- [ ] A list of IDs
- [ ] A string containing all of the tokens

Submit **You got all the answers!**

**10. Is there something wrong with the following code?**

```
from transformers import AutoTokenizer, AutoModel

tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
model = AutoModel.from_pretrained("gpt2")
```

End-of-chapter quiz
1. What is the order of the language modeling pipeline?
2. How many dimensions does the tensor output by the base Transformer model have, and what are they?
3. Which of the following is an example of subword tokenization?
4. What is a model head?
5. What is an AutoModel?
6. What are the techniques to be aware of when batching sequences of different lengths together?
7. What is the point of applying a SoftMax function to the logits output by a sequence classification model?
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10. Is there something wrong with the following code?



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**Correct!** Correct! That's not the only reason we use a SoftMax function, though.

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tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
model = AutoModel.from_pretrained("gpt2")

encoded = tokenizer("Hey!", return_tensors="pt")
result = model(**encoded)
```

- [ ] No, it seems correct.
- [x] The tokenizer and model should always be from the same checkpoint.

**Correct!** Right!

- [ ] It's good practice to pad and truncate with the tokenizer as every input is a batch.

Submit **You got all the answers!**

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← Optimized Inference Deployment

Complete Chapter

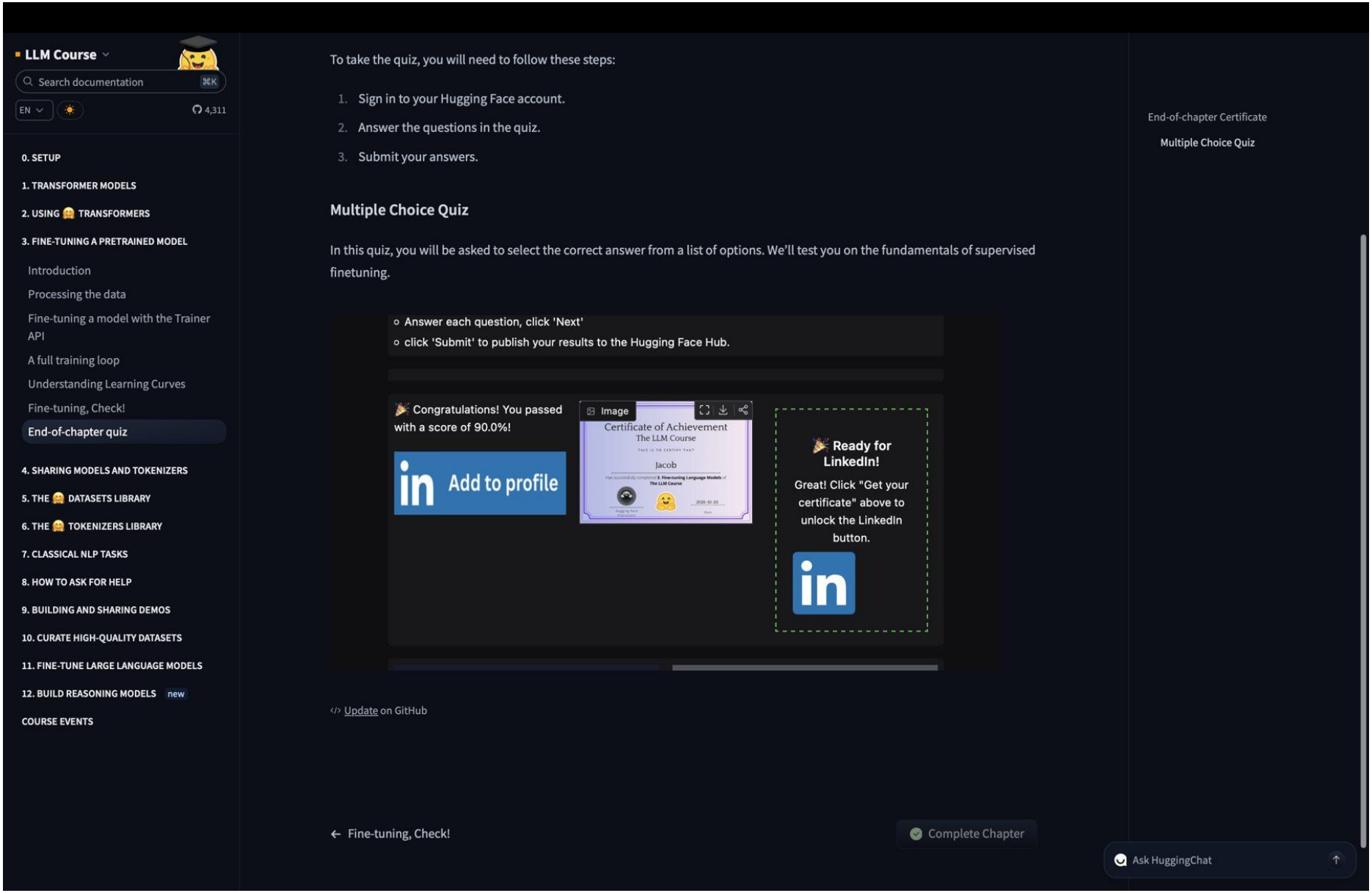
End-of-chapter quiz

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# Chapter 3 – Fine-tuning a Pretrained Model: Certification Exam (90%) and Certificate



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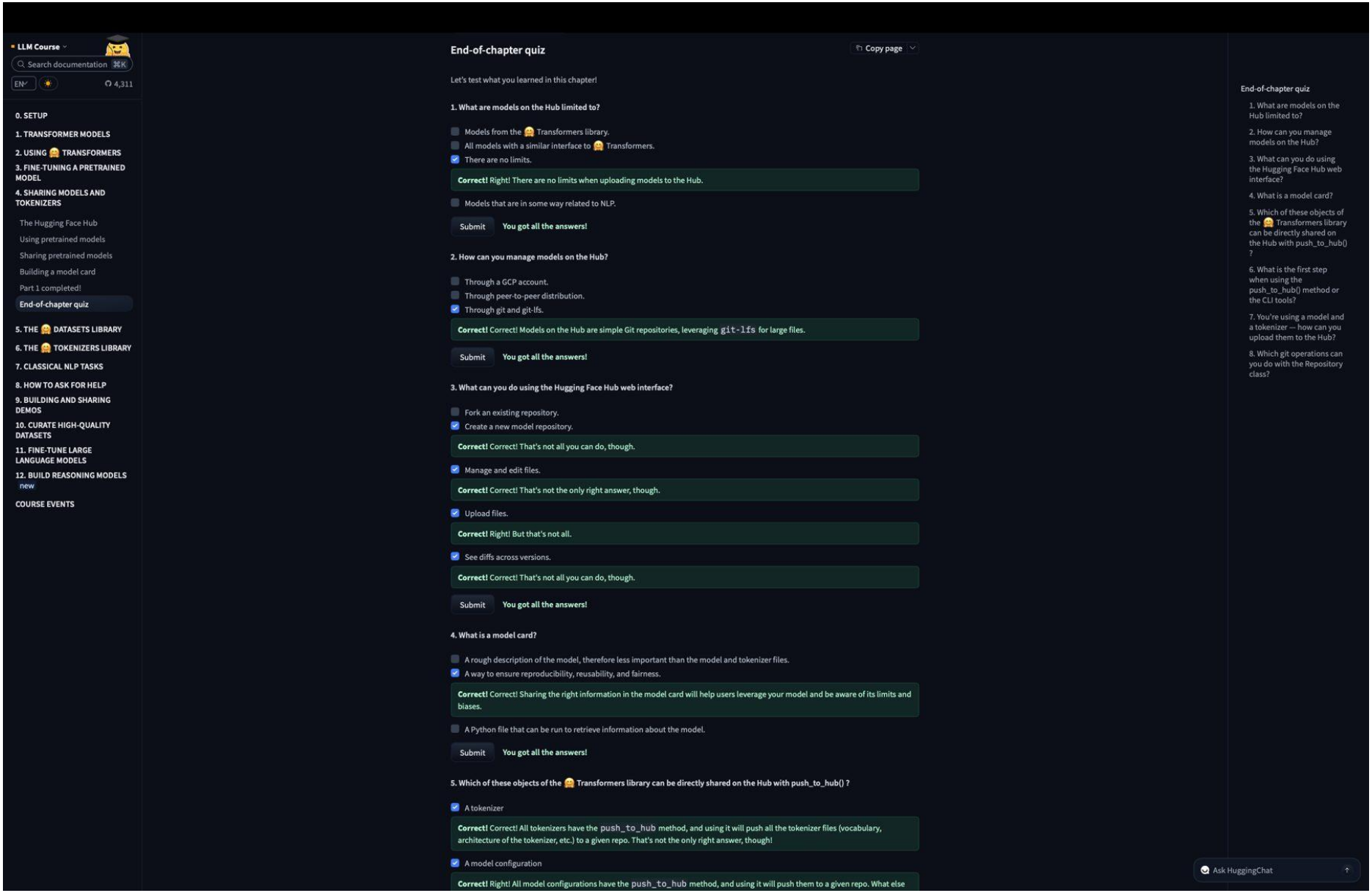
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# Chapter 4 – Sharing Models and Tokenizers: Quiz



**5. Which of these objects of the 🤗 Transformers library can be directly shared on the Hub with push_to_hub() ?**

- [x] A tokenizer

**Correct!** Correct! All tokenizers have the `push_to_hub` method, and using it will push all the tokenizer files (vocabulary, architecture of the tokenizer, etc.) to a given repo. That's not the only right answer, though!

- [x] A model configuration

**Correct!** Right! All model configurations have the `push_to_hub` method, and using it will push them to a given repo. What else can you share?

- [x] A model

**Correct!** Correct! All models have the `push_to_hub` method, and using it will push them and their configuration files to a given repo. That's not all you can share, though.

- [x] A Trainer

**Correct!** That's right — the `Trainer` also implements the `push_to_hub` method, and using it will upload the model, its configuration, the tokenizer, and a model card draft to a given repo. Try another answer!

Submit **You got all the answers!**

**6. What is the first step when using the push_to_hub() method or the CLI tools?**

- [ ] Log in on the website.
- [x] Run 'huggingface-cli login' in a terminal.

**Correct!** Correct — this will download and cache your personal token.

- [x] Run 'notebook_login()' in a notebook.

**Correct!** Correct — this will display a widget to let you authenticate.

Submit **You got all the answers!**

**7. You're using a model and a tokenizer — how can you upload them to the Hub?**

- [x] By calling the push_to_hub method directly on the model and the tokenizer.

**Correct!** Correct!

- [ ] Within the Python runtime, by wrapping them in a `huggingface_hub` utility.
- [ ] By saving them to disk and calling `transformers-cli upload-model`

Submit **You got all the answers!**

**8. Which git operations can you do with the Repository class?**

- [x] A commit.

**Correct!** Correct, the `git_commit()` method is there for that.

- [x] A pull

**Correct!** That is the purpose of the `git_pull()` method.

- [x] A push

**Correct!** The method `git_push()` does this.

- [ ] A merge

Submit **You got all the answers!**

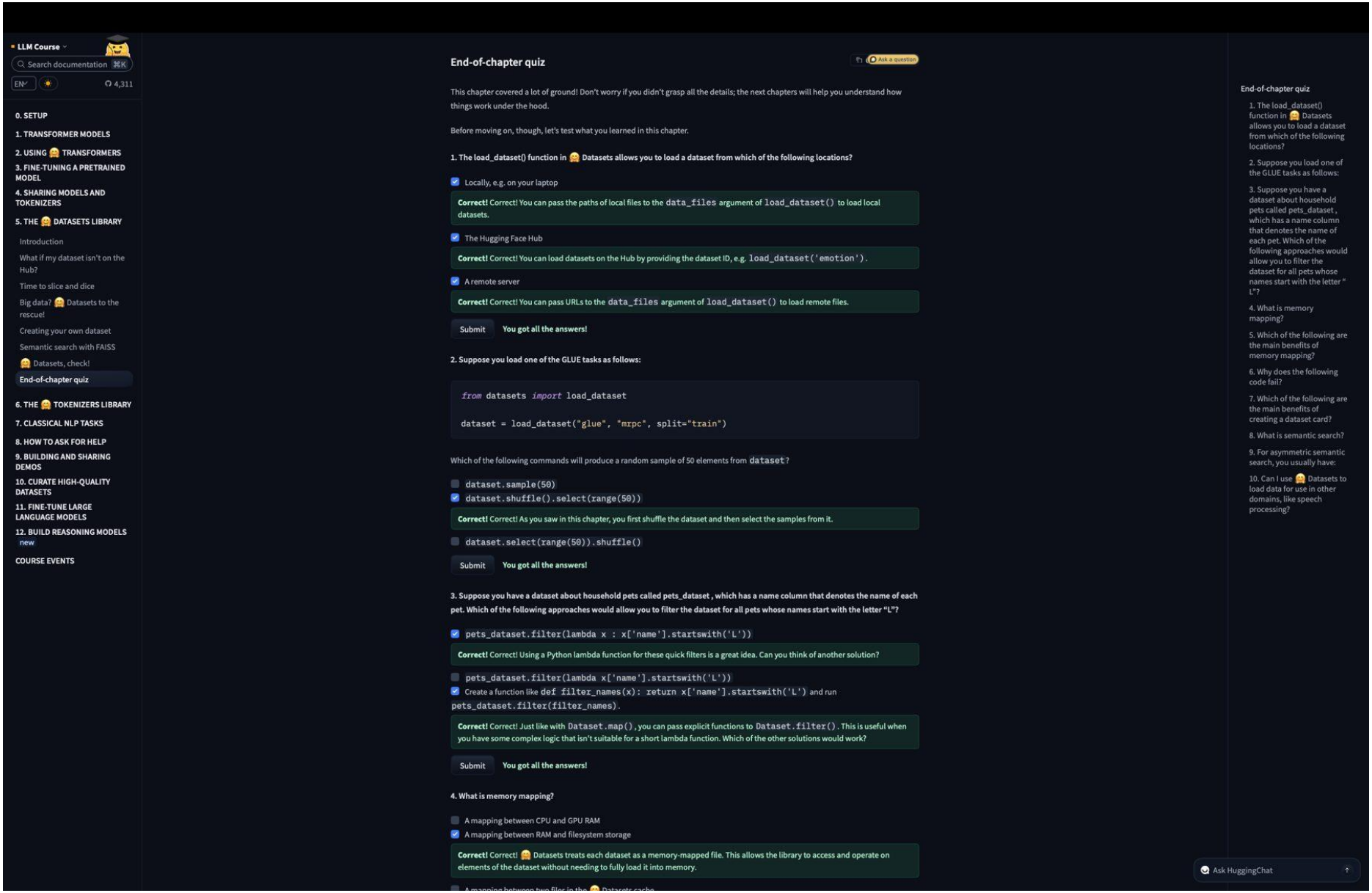
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End-of-chapter quiz
1. What are models on the Hub limited to?
2. How can you manage models on the Hub?
3. What can you do using the Hugging Face Hub web interface?
4. What is a model card?
5. Which of these objects of the 🤗 Transformers library can be directly shared on the Hub with push_to_hub() ?
6. What is the first step when using the push_to_hub() method or the CLI tools?
7. You're using a model and a tokenizer — how can you upload them to the Hub?
8. Which git operations can you do with the Repository class?



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# Chapter 5 – The Datasets Library: Quiz



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**4. What is memory mapping?**

- [ ] A mapping between CPU and GPU RAM
- [x] A mapping between RAM and filesystem storage

**Correct!** Correct! 🤗 Datasets treats each dataset as a memory-mapped file. This allows the library to access and operate on elements of the dataset without needing to fully load it into memory.

- [ ] A mapping between two files in the 🤗 Datasets cache

Submit **You got all the answers!**

**5. Which of the following are the main benefits of memory mapping?**

- [x] Accessing memory-mapped files is faster than reading from or writing to disk.

**Correct!** Correct! This allows 🤗 Datasets to be blazing fast. That's not the only benefit, though.

- [x] Applications can access segments of data in an extremely large file without having to read the whole file into RAM first.

**Correct!** Correct! This allows 🤗 Datasets to load multi-gigabyte datasets on your laptop without blowing up your CPU. What other advantage does memory mapping offer?

- [ ] It consumes less energy, so your battery lasts longer.

Submit **You got all the answers!**

**6. Why does the following code fail?**

```
from datasets import load_dataset

dataset = load_dataset("allocine", streaming=True, split="train")
dataset[0]
```

- [ ] It tries to stream a dataset that's too large to fit in RAM.
- [x] It tries to access an `IterableDataset`.

**Correct!** Correct! An `IterableDataset` is a generator, not a container, so you should access its elements using `next(iter(dataset))`.

- [ ] The `allocine` dataset doesn't have a `train` split.

Submit **You got all the answers!**

**7. Which of the following are the main benefits of creating a dataset card?**

- [x] It provides information about the intended use and supported tasks of the dataset so others in the community can make an informed decision about using it.

**Correct!** Correct! Undocumented datasets may be used to train models that may not reflect the intentions of the dataset creators, or may produce models whose legal status is murky if they're trained on data that violates privacy or licensing restrictions. This isn't the only benefit, though!

- [x] It helps draw attention to the biases that are present in a corpus.

**Correct!** Correct! Almost all datasets have some form of bias, which can produce negative consequences downstream. Being aware of them helps model builders understand how to address the inherent biases. What else do dataset cards help with?

- [x] It improves the chances that others in the community will use my dataset.

**Correct!** Correct! A well-written dataset card will tend to lead to higher usage of your precious dataset. What other benefits does it offer?

Submit **You got all the answers!**

**8. What is semantic search?**

- [ ] A way to search for exact matches between the words in a query and the documents in a corpus

End-of-chapter quiz
1. The load_dataset() function in 🤗 Datasets allows you to load a dataset from which of the following locations?
2. Suppose you load one of the GLUE tasks as follows:
3. Suppose you have a dataset about household pets called pets_dataset , which has a name column that denotes the name of each pet. Which of the following approaches would allow you to filter the dataset for all pets whose names start with the letter "L"?
4. What is memory mapping?
5. Which of the following are the main benefits of memory mapping?
6. Why does the following code fail?
7. Which of the following are the main benefits of creating a dataset card?
8. What is semantic search?
9. For asymmetric semantic search, you usually have:
10. Can I use 🤗 Datasets to load data for use in other domains, like speech processing?



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**8. What is semantic search?**

- [ ] A way to search for exact matches between the words in a query and the documents in a corpus
- [x] A way to search for matching documents by understanding the contextual meaning of a query

**Correct!** Correct! Semantic search uses embedding vectors to represent queries and documents, and uses a similarity metric to measure the amount of overlap between them. How else might you describe it?

- [x] A way to improve search accuracy

**Correct!** Correct! Semantic search engines can capture the intent of a query much better than keyword matching and typically retrieve documents with higher precision. But this isn't the only right answer - what else does semantic search provide?

Submit **You got all the answers!**

**9. For asymmetric semantic search, you usually have:**

- [x] A short query and a longer paragraph that answers the query

**Correct!** Correct!

- [ ] Queries and paragraphs that are of about the same length
- [ ] A long query and a shorter paragraph that answers the query

Submit **You got all the answers!**

**10. Can I use 🤗 Datasets to load data for use in other domains, like speech processing?**

- [ ] No
- [x] Yes

**Correct!** Correct! Check out the exciting developments with speech and vision in the 🤗 Transformers library to see how 🤗 Datasets is used in these domains.

Submit **You got all the answers!**

</> Update on GitHub

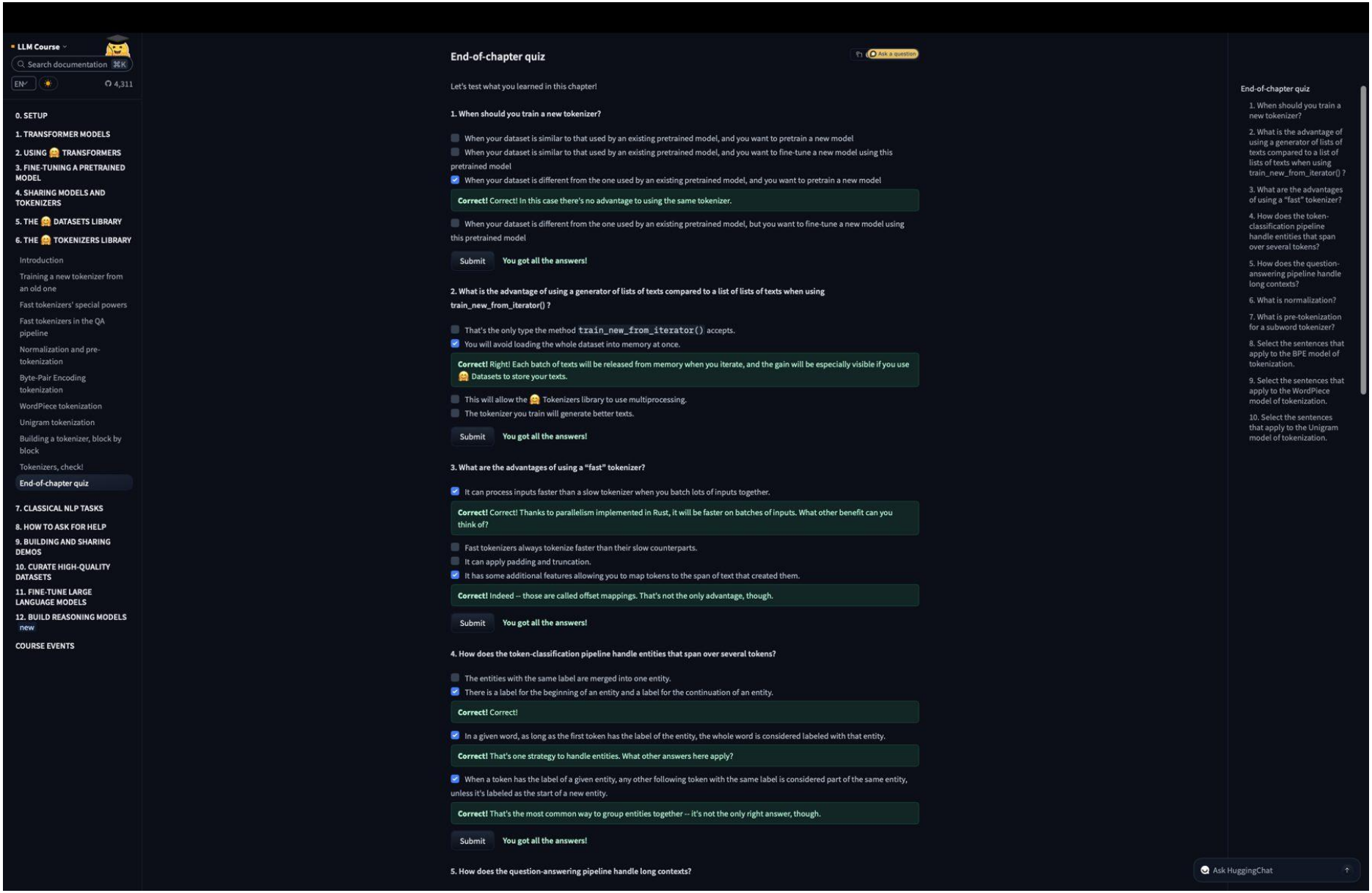
← 🤗 Datasets, check!
Complete Chapter

End-of-chapter quiz
1. The load_dataset() function in 🤗 Datasets allows you to load a dataset from which of the following locations?
2. Suppose you load one of the GLUE tasks as follows:
3. Suppose you have a dataset about household pets called pets_dataset , which has a name column that denotes the name of each pet. Which of the following approaches would allow you to filter the dataset for all pets whose names start with the letter " L"?
4. What is memory mapping?
5. Which of the following are the main benefits of memory mapping?
6. Why does the following code fail?
7. Which of the following are the main benefits of creating a dataset card?
8. What is semantic search?
9. For asymmetric semantic search, you usually have:
10. Can I use 🤗 Datasets to load data for use in other domains, like speech processing?



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# Chapter 6 – The Tokenizers Library: Quiz



0. SETUP
1. TRANSFORMER MODELS
2. USING 🤗 TRANSFORMERS
3. FINE-TUNING A PRETRAINED MODEL
4. SHARING MODELS AND TOKENIZERS
5. THE 🤗 DATASETS LIBRARY
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Fast tokenizers in the QA pipeline
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Byte-Pair Encoding tokenization
WordPiece tokenization
Unigram tokenization
Building a tokenizer, block by block
Tokenizers, check!
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11. FINE-TUNE LARGE LANGUAGE MODELS
12. BUILD REASONING MODELS new
COURSE EVENTS

**5. How does the question-answering pipeline handle long contexts?**

- [ ] It doesn't really, as it truncates the long context at the maximum length accepted by the model.
- [ ] It splits the context into several parts and averages the results obtained.
- [x] It splits the context into several parts (with overlap) and finds the maximum score for an answer in each part.

**Correct!** That's the correct answer!

- [ ] It splits the context into several parts (without overlap, for efficiency) and finds the maximum score for an answer in each part.

Submit **You got all the answers!**

**6. What is normalization?**

- [x] It's any cleanup the tokenizer performs on the texts in the initial stages.

**Correct!** That's correct -- for instance, it might involve removing accents or whitespace, or lowercasing the inputs.

- [ ] It's a data augmentation technique that involves making the text more normal by removing rare words.
- [ ] It's the final post-processing step where the tokenizer adds the special tokens.
- [ ] It's when the embeddings are made with mean 0 and standard deviation 1, by subtracting the mean and dividing by the std.

Submit **You got all the answers!**

**7. What is pre-tokenization for a subword tokenizer?**

- [ ] It's the step before the tokenization, where data augmentation (like random masking) is applied.
- [ ] It's the step before the tokenization, where the desired cleanup operations are applied to the text.
- [x] It's the step before the tokenizer model is applied, to split the input into words.

**Correct!** That's the correct answer!

- [ ] It's the step before the tokenizer model is applied, to split the input into tokens.

Submit **You got all the answers!**

**8. Select the sentences that apply to the BPE model of tokenization.**

- [x] BPE is a subword tokenization algorithm that starts with a small vocabulary and learns merge rules.

**Correct!** That's the case indeed!

- [ ] BPE is a subword tokenization algorithm that starts with a big vocabulary and progressively removes tokens from it.
- [x] BPE tokenizers learn merge rules by merging the pair of tokens that is the most frequent.

**Correct!** That's correct!

- [ ] A BPE tokenizer learns a merge rule by merging the pair of tokens that maximizes a score that privileges frequent pairs with less frequent individual parts.
- [x] BPE tokenizes words into subwords by splitting them into characters and then applying the merge rules.

**Correct!** That's correct!

- [ ] BPE tokenizes words into subwords by finding the longest subword starting from the beginning that is in the vocabulary, then repeating the process for the rest of the text.

Submit **You got all the answers!**

**9. Select the sentences that apply to the WordPiece model of tokenization.**

- [x] WordPiece is a subword tokenization algorithm that starts with a small vocabulary and learns merge rules.

**Correct!** That's the case indeed!

- [ ] WordPiece is a subword tokenization algorithm that starts with a big vocabulary and progressively removes tokens from it.
- [ ] WordPiece tokenizers learn merge rules by merging the pair of tokens that is the most frequent.
- [x] A WordPiece tokenizer learns a merge rule by merging the pair of tokens that maximizes a score that privileges frequent pairs with less frequent individual parts.

End-of-chapter quiz
1. When should you train a new tokenizer?
2. What is the advantage of using a generator of lists of texts compared to a list of lists of texts when using train_new_from_iterator() ?
3. What are the advantages of using a "fast" tokenizer?
4. How does the token-classification pipeline handle entities that span over several tokens?
5. How does the question-answering pipeline handle long contexts?
6. What is normalization?
7. What is pre-tokenization for a subword tokenizer?
8. Select the sentences that apply to the BPE model of tokenization.
9. Select the sentences that apply to the WordPiece model of tokenization.
10. Select the sentences that apply to the Unigram model of tokenization.



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12. BUILD REASONING MODELS new
COURSE EVENTS

**Correct!** That's the case indeed!

- [ ] BPE is a subword tokenization algorithm that starts with a big vocabulary and progressively removes tokens from it.
- [x] BPE tokenizers learn merge rules by merging the pair of tokens that is the most frequent.

**Correct!** That's correct!

- [ ] A BPE tokenizer learns a merge rule by merging the pair of tokens that maximizes a score that privileges frequent pairs with less frequent individual parts.
- [x] BPE tokenizes words into subwords by splitting them into characters and then applying the merge rules.

**Correct!** That's correct!

- [ ] BPE tokenizes words into subwords by finding the longest subword starting from the beginning that is in the vocabulary, then repeating the process for the rest of the text.

Submit **You got all the answers!**

**9. Select the sentences that apply to the WordPiece model of tokenization.**

- [x] WordPiece is a subword tokenization algorithm that starts with a small vocabulary and learns merge rules.

**Correct!** That's the case indeed!

- [ ] WordPiece is a subword tokenization algorithm that starts with a big vocabulary and progressively removes tokens from it.
- [ ] WordPiece tokenizers learn merge rules by merging the pair of tokens that is the most frequent.
- [x] A WordPiece tokenizer learns a merge rule by merging the pair of tokens that maximizes a score that privileges frequent pairs with less frequent individual parts.

**Correct!** That's correct!

- [ ] WordPiece tokenizes words into subwords by finding the most likely segmentation into tokens, according to the model.
- [x] WordPiece tokenizes words into subwords by finding the longest subword starting from the beginning that is in the vocabulary, then repeating the process for the rest of the text.

**Correct!** Yes, this is how WordPiece proceeds for the encoding.

Submit **You got all the answers!**

**10. Select the sentences that apply to the Unigram model of tokenization.**

- [ ] Unigram is a subword tokenization algorithm that starts with a small vocabulary and learns merge rules.
- [x] Unigram is a subword tokenization algorithm that starts with a big vocabulary and progressively removes tokens from it.

**Correct!** That's correct!

- [x] Unigram adapts its vocabulary by minimizing a loss computed over the whole corpus.

**Correct!** That's correct!

- [ ] Unigram adapts its vocabulary by keeping the most frequent subwords.
- [x] Unigram tokenizes words into subwords by finding the most likely segmentation into tokens, according to the model.

**Correct!** That's correct!

- [ ] Unigram tokenizes words into subwords by splitting them into characters, then applying the merge rules.

Submit **You got all the answers!**

Update on GitHub

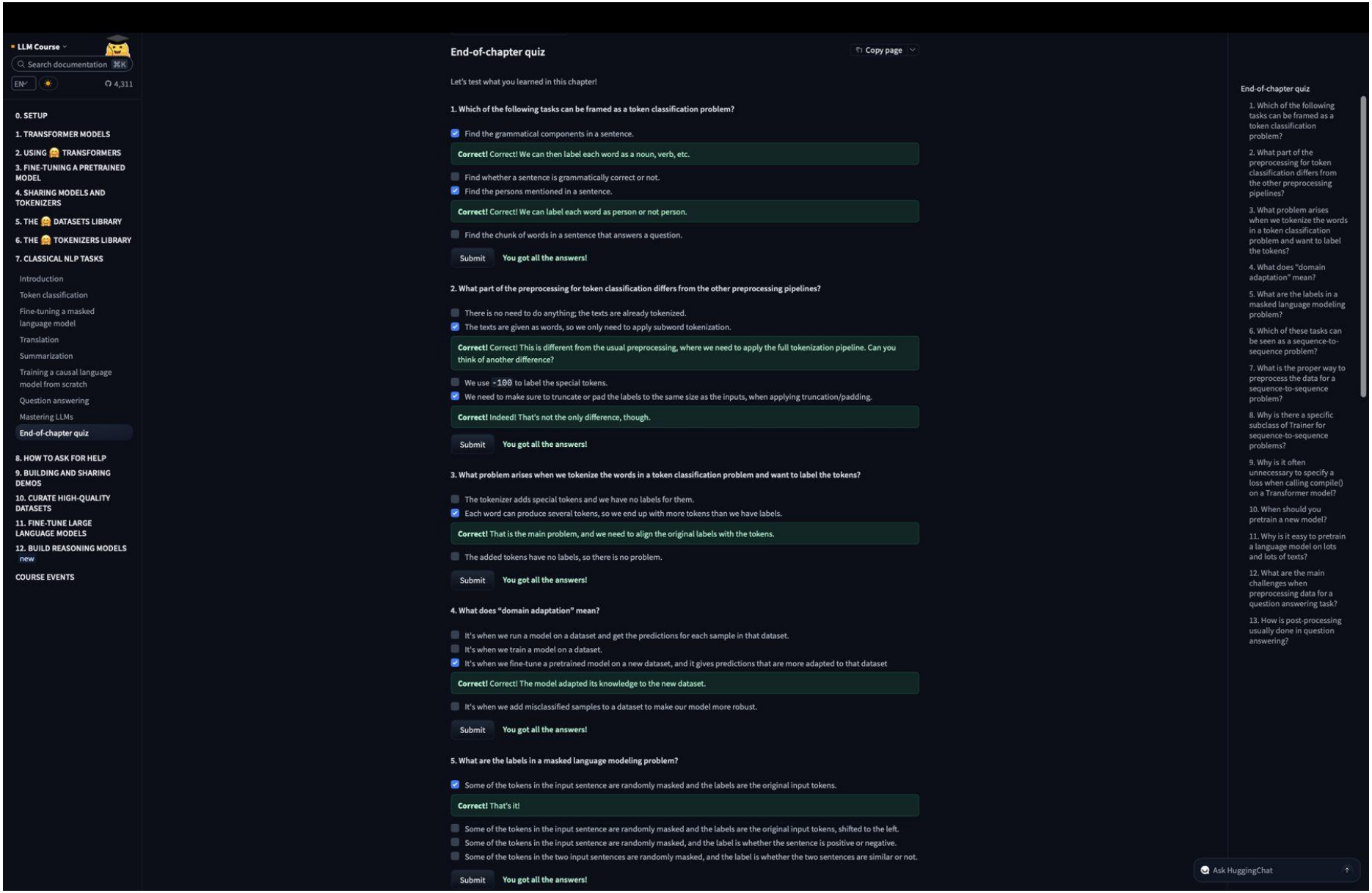
← Tokenizers, check!
Complete Chapter

End-of-chapter quiz
1. When should you train a new tokenizer?
2. What is the advantage of using a generator of lists of texts compared to a list of lists of texts when using train_new_from_iterator() ?
3. What are the advantages of using a "fast" tokenizer?
4. How does the token-classification pipeline handle entities that span over several tokens?
5. How does the question-answering pipeline handle long contexts?
6. What is normalization?
7. What is pre-tokenization for a subword tokenizer?
8. **Select the sentences that apply to the BPE model of tokenization.**
9. Select the sentences that apply to the WordPiece model of tokenization.
10. Select the sentences that apply to the Unigram model of tokenization.



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# Chapter 7 – Classical NLP Tasks: Quiz



0. SETUP
1. TRANSFORMER MODELS
2. USING 🤗 TRANSFORMERS
3. FINE-TUNING A PRETRAINED MODEL
4. SHARING MODELS AND TOKENIZERS
5. THE 🤗 DATASETS LIBRARY
6. THE 🤗 TOKENIZERS LIBRARY
7. CLASSICAL NLP TASKS
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Fine-tuning a masked language model
Translation
Summarization
Training a causal language model from scratch
Question answering
Mastering LLMs
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8. HOW TO ASK FOR HELP
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11. FINE-TUNE LARGE LANGUAGE MODELS
12. BUILD REASONING MODELS new
COURSE EVENTS

**5. What are the labels in a masked language modeling problem?**

- [x] Some of the tokens in the input sentence are randomly masked and the labels are the original input tokens.

**Correct!** That's it!

- [ ] Some of the tokens in the input sentence are randomly masked and the labels are the original input tokens, shifted to the left.
- [ ] Some of the tokens in the input sentence are randomly masked, and the label is whether the sentence is positive or negative.
- [ ] Some of the tokens in the two input sentences are randomly masked, and the label is whether the two sentences are similar or not.

Submit **You got all the answers!**

**6. Which of these tasks can be seen as a sequence-to-sequence problem?**

- [x] Writing short reviews of long documents

**Correct!** Yes, that's a summarization problem. Try another answer!

- [x] Answering questions about a document

**Correct!** This can be framed as a sequence-to-sequence problem. It's not the only right answer, though.

- [x] Translating a text in Chinese into English

**Correct!** That's definitely a sequence-to-sequence problem. Can you spot another one?

- [x] Fixing the messages sent by my nephew/friend so they're in proper English

**Correct!** That's a kind of translation problem, so definitely a sequence-to-sequence task. This isn't the only right answer, though!

Submit **You got all the answers!**

**7. What is the proper way to preprocess the data for a sequence-to-sequence problem?**

- [ ] The inputs and targets have to be sent together to the tokenizer with `inputs=...` and `targets=...`.
- [ ] The inputs and the targets both have to be preprocessed, in two separate calls to the tokenizer.
- [ ] As usual, we just have to tokenize the inputs.
- [x] The inputs have to be sent to the tokenizer, and the targets too, but under a special context manager.

**Correct!** That's correct, the tokenizer needs to be put into target mode by that context manager.

Submit **You got all the answers!**

**8. Why is there a specific subclass of Trainer for sequence-to-sequence problems?**

- [ ] Because sequence-to-sequence problems use a custom loss, to ignore the labels set to `-100`
- [x] Because sequence-to-sequence problems require a special evaluation loop

**Correct!** That's correct. Sequence-to-sequence models' predictions are often run using the `generate()` method.

- [ ] Because the targets are texts in sequence-to-sequence problems
- [ ] Because we use two models in sequence-to-sequence problems

Submit **You got all the answers!**

**10. When should you pretrain a new model?**

- [x] When there is no pretrained model available for your specific language

**Correct!** That's correct.

- [ ] When you have lots of data available, even if there is a pretrained model that could work on it
- [x] When you have concerns about the bias of the pretrained model you are using

**Correct!** That is true, but you have to make very sure the data you will use for training is really better.

- [ ] When the pretrained models available are just not good enough

Submit **You got all the answers!**

End-of-chapter quiz
1. Which of the following tasks can be framed as a token classification problem?
2. What part of the preprocessing for token classification differs from the other preprocessing pipelines?
3. What problem arises when we tokenize the words in a token classification problem and want to label the tokens?
4. What does "domain adaptation" mean?
5. What are the labels in a masked language modeling problem?
6. Which of these tasks can be seen as a sequence-to-sequence problem?
7. What is the proper way to preprocess the data for a sequence-to-sequence problem?
8. Why is there a specific subclass of Trainer for sequence-to-sequence problems?
9. Why is it often unnecessary to specify a loss when calling compile() on a Transformer model?
10. When should you pretrain a new model?
11. Why is it easy to pretrain a language model on lots and lots of texts?
12. What are the main challenges when preprocessing data for a question answering task?
13. How is post-processing usually done in question answering?

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- [ ] Because we use two models in sequence-to-sequence problems

Submit **You got all the answers!**

**10. When should you pretrain a new model?**

- [x] When there is no pretrained model available for your specific language

**Correct!** That's correct.

- [ ] When you have lots of data available, even if there is a pretrained model that could work on it
- [x] When you have concerns about the bias of the pretrained model you are using

**Correct!** That is true, but you have to make very sure the data you will use for training is really better.

- [ ] When the pretrained models available are just not good enough

Submit **You got all the answers!**

**11. Why is it easy to pretrain a language model on lots and lots of texts?**

- [ ] Because there are plenty of texts available on the internet
- [x] Because the pretraining objective does not require humans to label the data

**Correct!** That's correct, language modeling is a self-supervised problem.

- [ ] Because the 🤗 Transformers library only requires a few lines of code to start the training

Submit **You got all the answers!**

**12. What are the main challenges when preprocessing data for a question answering task?**

- [ ] You need to tokenize the inputs.
- [x] You need to deal with very long contexts, which give several training features that may or may not have the answer in them.

**Correct!** This is definitely one of the challenges.

- [ ] You need to tokenize the answers to the question as well as the inputs.
- [x] From the answer span in the text, you have to find the start and end token in the tokenized input.

**Correct!** That's one of the hard parts, yes!

Submit **You got all the answers!**

**13. How is post-processing usually done in question answering?**

- [ ] The model gives you the start and end positions of the answer, and you just have to decode the corresponding span of tokens.
- [ ] The model gives you the start and end positions of the answer for each feature created by one example, and you just have to decode the corresponding span of tokens in the one that has the best score.
- [x] The model gives you the start and end positions of the answer for each feature created by one example, and you just have to match them to the span in the context for the one that has the best score.

**Correct!** That's it in a nutshell!

- [ ] The model generates an answer, and you just have to decode it.

Submit **You got all the answers!**

</> Update on GitHub

← Mastering LLMs
Complete Chapter

End-of-chapter quiz
1. Which of the following tasks can be framed as a token classification problem?
2. What part of the preprocessing for token classification differs from the other preprocessing pipelines?
3. What problem arises when we tokenize the words in a token classification problem and want to label the tokens?
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